Exercise – Understanding maximum simulated likelihood estimation

1. Aim of the exercise

This exercise estimates a random-coefficient model using maximum simulated likelihood (MSL). Whereas standard mixed-logit applications are typically reduced-form, the specification here is grounded in an explicit life-cycle economic model, demonstrating how random-coefficient methods can be applied within a more flexible structural framework.

Implementation of MSL builds on econometric concepts covered in earlier exercises in *Simulation Methods in Econometrics: Theory and Applications in MATLAB*. These include Monte Carlo integration, quasi-random sampling, and the construction of simulated likelihood functions. Readers are therefore advised to review those materials before proceeding.

This exercise also assumes familiarity with numerical optimization in MATLAB. The MATLAB workshop in the book explains how to set up and minimize objective functions using `fmincon`, including the use of function handles, initial guesses, and solver options. We do not repeat those instructions here; the reader is expected to consult the workshop for the optimization background needed to implement the estimation routine below.

2. Data

The life-cycle hypothesis implies that older individuals prefer a gradual transition from full-time work to full retirement, smoothing both consumption and leisure over time. Partial-retirement schemes provide a natural mechanism for such smoothing. However, actual participation remains low. A plausible explanation is that many workers cannot supplement reduced part-time earnings with personal savings or partial pension benefits. In addition, institutional features of pension systems and labor contracts often restrict flexible claiming or late-career reductions in working hours, limiting the practical applicability of partial-retirement arrangements.

Because few individuals actually participate in partial retirement, revealed-preference data offer limited scope for studying such behavior. Kantarcı et al. (2025) address this limitation by eliciting retirement preferences through a stated-choice survey. The survey was fielded in 2017 in the Longitudinal Internet Studies for the Social Sciences (LISS) panel and consists of two main components. The first collects background characteristics and information on respondents' work lives. The second is designed to elicit preferences for abrupt versus partial retirement. In this part, respondents answer a series of stated-choice questions in which they trade off working more hours in exchange for a higher pension against working fewer hours with a lower pension.

Figure 1 shows an example stated-choice question. It starts with a short introduction and then briefly describes three alternative retirement scenarios, followed by a timeline giving the number of hours worked and the earnings and pension income at each age. Respondents are asked to choose their favorite retirement scenario among the three, based on their own preferences.

To increase variation in choices, the question is asked to a given respondent for a total of three times, changing the ages of partial and full retirement in each case. The pension income is adjusted for working more or fewer years. The three versions use age 61, age 63 and 65 as the age of early or partial retirement, and age 66, age 68 and 70 as the age of late retirement, respectively. For ease of reference, we denote the three question variants – corresponding to early/partial retirement at ages 61, 63, and 65 – as retirement-age regimes 61, 63, and 65.

The duration of partial retirement is randomized between four and five years across two sub-samples of respondents. The survey also includes additional stated-choice questions. For the present exercise, the sample is restricted to respondents assigned to the five-year partial-retirement scenario, and the supplementary questions are excluded. These restrictions are made to simplify the exercise and they do not alter the core estimation problem.

3. Random utility model

To model the stated retirement preferences, Kantarcı et al. (2025) utilize a life-cycle framework. Assume that the total utility, U_i^q, of retirement scenario q for individual $i = 1, \dots, n$ has the following form:

$$U_i^q = \sum_{t=60}^{100} \rho^{(t-60)} \pi_t U_{it}^q \tag{1}$$

where ρ is the discount factor. π_t is the probability of surviving from one age to the next, given survival up to age 60. U_{it}^q is the utility at age $t = 60, \dots, 100$. The time horizon is fixed at 100 years of age. q is an early abrupt retirement scenario (E), a partial retirement scenario (P), or a late abrupt retirement scenario (L). In all trajectories, the agent is working full-time at age 60. At later ages t, leisure l_{it}^q and net income y_{it}^q vary across trajectories.

Within period utility is specified as follows:

$$U_{it}^q = \alpha_{it}^l \ln(l_{it}^q) + \alpha^y \ln(y_{it}^q) + \alpha^{ly} \ln(l_{it}^q) \ln(y_{it}^q) \tag{2}$$

$$\alpha_{it}^l = X_i \beta^l + \eta^l t + e_i^l \tag{3}$$

$$e_i^l \sim N\left(0, \sigma_l^2\right) \text{ and } e_i^l \text{ independent of } X_i \tag{4}$$

$$l_{it}^q = T - h_{it}^q \tag{5}$$

where T is the number of hours available for work and leisure in a working week and is a parameter to be estimated. h_{it}^q denotes hours of paid work per week. At each age t, the person can work full-time ($h_{it}^q = 40$), can be partially retired (h_{it}^q =20), or can be fully retired ($h_{it}^q = 0$).

During full-time employment, net income y_{it}^q consists of after-tax earnings. During full retirement, it corresponds to after-tax pension income. During partial retirement, y_{it}^q includes both part-time earnings and partial pension income.

The preference parameters α_{it}^l and α^{ly} drive the marginal utility of leisure time for individual i at age t. α_{it}^l depends on observed characteristics X_i such as age, gender and home ownership, and, through e_i^l, on individual i's unobserved characteristics. The effect of age t is captured by $\eta^l t$. We expect $\eta^l > 0$, since individuals' valuation of leisure will typically increase with age. A nonzero parameter α^{ly} implies that the marginal utility of leisure also varies with income.

Together with α^{ly}, the parameter α^y determines the marginal utility of income. Both parameters are treated as constants, to avoid multicollinearity and imprecise estimates.

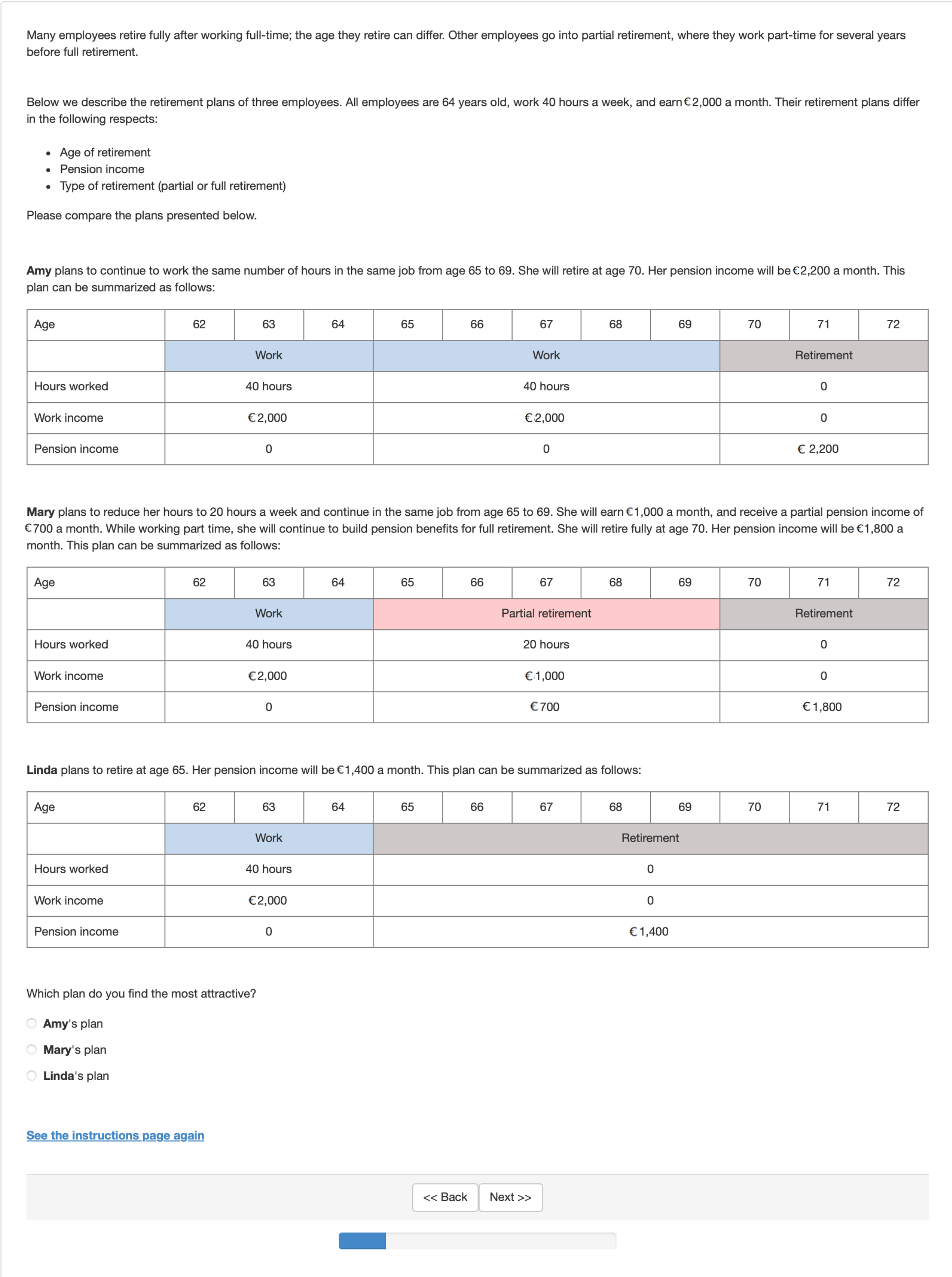
Introducing error terms u_i^q as in a standard random utility model, the model takes the following form:

$$V_i^q = U_i^q + u_i^q \tag{6}$$

$$u_i^q \sim \text{i.i.d. type I extreme value and independent of } X_i, e_i^l \tag{7}$$

$$F(u_i^q) = e^{-e^{-u_i^q}} \tag{8}$$

where F denotes the cumulative distribution function.



Many employees retire fully after working full-time; the age they retire can differ. Other employees go into partial retirement, where they work part-time for several years before full retirement.

Below we describe the retirement plans of three employees. All employees are 64 years old, work 40 hours a week, and earn €2,000 a month. Their retirement plans differ in the following respects:

- Age of retirement
- Pension income
- Type of retirement (partial or full retirement)

Please compare the plans presented below.

Amy plans to continue to work the same number of hours in the same job from age 65 to 69. She will retire at age 70. Her pension income will be €2,200 a month. This plan can be summarized as follows:

Age	62	63	64	65	66	67	68	69	70	71	72
	Work			Work					Retirement		
Hours worked	40 hours			40 hours					0		
Work income	€2,000			€2,000					0		
Pension income	0			0					€ 2,200		

Mary plans to reduce her hours to 20 hours a week and continue in the same job from age 65 to 69. She will earn €1,000 a month, and receive a partial pension income of €700 a month. While working part time, she will continue to build pension benefits for full retirement. She will retire fully at age 70. Her pension income will be €1,800 a month. This plan can be summarized as follows:

Age	62	63	64	65	66	67	68	69	70	71	72
	Work			Partial retirement					Retirement		
Hours worked	40 hours			20 hours					0		
Work income	€2,000			€1,000					0		
Pension income	0			€700					€1,800		

Linda plans to retire at age 65. Her pension income will be €1,400 a month. This plan can be summarized as follows:

Age	62	63	64	65	66	67	68	69	70	71	72
	Work			Retirement							
Hours worked	40 hours			0							
Work income	€2,000			0							
Pension income	0			€1,400							

Which plan do you find the most attractive?

- **Amy**'s plan
- **Mary**'s plan
- **Linda**'s plan

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Figure 1: The stated preference question asking to choose among early, partial and late retirement

The observed choice in question Q is given by

$$C_i^Q = q \text{ if } V_i^q > V_i^p \text{ for all } p \neq q. \tag{9}$$

As described earlier, as we consider in the present exercise, respondents choose among retirement scenarios in three questions.

Define $u_i^q - u_i^p \equiv u_i^{qp}$. The assumptions on u_i^q imply that u_i^{qp} has a standard logistic distribution and that the probability of choosing scenario q among alternative scenarios j in question Q, conditional on all individual and scenario characteristics and preference parameters, including the unobserved preference term e_i^l, is given by

$$\mathrm{P}\left(C_i^Q = q \,\middle|\, A_i, e_i^l\right) = \frac{e^{U_i^q}}{\sum\limits_j e^{U_i^j}} \tag{10}$$

where $A_i = \{l_{it}^q, y_{it}^q, X_i, t, \beta^l, \eta^l, \alpha^y, \alpha^{ly}\}$ is the set of all relevant individual and scenario characteristics and parameters.

Because the individual preference parameter e_i^l is unobserved, we cannot directly use these conditional probabilities to construct the sample likelihood. Instead, to remove the dependence on this unobserved individual effect, we integrate the product of the conditional probabilities over the assumed normal population distribution of e_i^l. This yields the unconditional likelihood contribution for individual i:

$$L_i = \int\limits_{-\infty}^{\infty} \prod_{Q=1}^{K(i)} \mathrm{P}\left(C_i^Q = q \,\middle|\, A_i, e_i^l\right) \frac{1}{\sigma_l} \phi\left(\frac{e_i^l}{\sigma_l}\right) \mathrm{d}e_i^l \tag{11}$$

where $K(i)$ is the number of questions answered by respondent i (which is 3 in the current exercise). Here, the general normal density of e_i^l is written in terms of the standard normal density function, $\phi(\cdot)$. This formulation is standard in mixed logit models because expressing the density via the standardized variable $\frac{e_i^l}{\sigma_l}$, and scaling by $\frac{1}{\sigma_l}$ to preserve a total probability of one, allows the model to be estimated computationally using a fixed set of standard normal random draws.

This integral cannot be solved analytically because the integrand is a non-linear combination of a logistic choice probability and a normal density function, a combination that lacks a closed-form antiderivative. In fact, the name "mixed logit" stems from this continuous mixture of distributions. We therefore evaluate it numerically using Monte Carlo integration. Under this approach, the continuous integral is approximated by simulating the distribution of e_i^l through random draws and calculating the simulated sample average:

$$\hat{L}_i = \frac{1}{S} \sum_{s=1}^{S} \prod_{Q=1}^{K(i)} \mathrm{P}\left(C_i^Q = q \,\middle|\, A_i, e_{i,s}^l\right) \tag{12}$$

where S is the number of simulations, and the simulated preference term is defined as

$$e_{i,s}^l = \sigma_l z_{i,s} \tag{13}$$

where $z_{i,s}$ are independent draws from a standard normal distribution.

The individual log-likelihood contribution is then

$$\ell_i = \log \hat{L}_i, \tag{14}$$

which transforms the simulated likelihood in equation (12) into a numerically stable additive form.

Summing these individual contributions over all individuals yields the sample log-likelihood,

$$\ell(\theta) = \sum_{i=1}^{N} \ell_i, \tag{15}$$

which is maximized with respect to the structural parameters to obtain the Simulated Maximum Likelihood estimator (Gouriéroux and Monfort, 1990).

In this setting, a large number of pseudo-random draws is typically required to minimize the simulation bias introduced by the logarithmic transformation of the simulated likelihood. The number of draws, and hence the estimation time, can be significantly reduced (while maintaining the same simulation variance) by replacing pseudo-random draws with quasi-random numbers from a Halton sequence. Estimates of the covariance matrix of the parameter estimates are based upon the asymptotic results established by Gouriéroux and Monfort.

While the one-dimensional integral in equation (11) can be evaluated using numerical quadrature or other deterministic methods, simulated maximum likelihood is standard in mixed logit applications and easily accommodates the random-coefficient structure. The use of quasi-random Halton draws further improves numerical efficiency relative to pseudo-random simulation methods.

4. Implementation

The MSL estimation procedure that we implement in MATLAB consists of four script files — `load_data.m`, `generate_halton_draws.m`, `likelihood_estimation.m`, and `results_table.m` — and two function files — `halton.m` and `likelihood.m`. The `halton.m` function is developed and explained in the quasi-random sampling exercise. The remaining files are described below.

4.1. Load the data

We begin by loading the data into the Workspace:

```
load('data.mat')
```

Once the data are loaded, the Workspace contains the following variables. `id` contains the identification numbers of respondents.

`X_i` is the matrix of observed characteristics, including variables such as age, gender, and home ownership.

`t` is a row vector with 41 values, ranging from ages 60 to 100. It indexes the age of the hypothetical individual as defined in equation (1).

The variable `choice_at_age_regime_61` records which of the three retirement scenarios a respondent selects in the question corresponding to retirement-age regime 61. A value of 1 indicates early (abrupt) retirement, 2 indicates partial retirement, and 3 indicates late (abrupt) retirement. The variables `choice_at_age_regime_63` and `choice_at_age_regime_65` contain the analogous choices for the questions featuring retirement-age regimes 63 and 65, respectively.

The variable `h_i_t_65E` contains, in each row, the number of hours worked per week at each age from 60 to 100 for a given individual, for the retirement scenario in which early retirement begins at age 65. This scenario is illustrated at the bottom of Figure 1: the hypothetical individual works 40 hours per week from ages 60 to 64 and retires at age 65, after which weekly hours drop to zero. Accordingly, the first five columns of `h_i_t_65E` take the value 40 (for ages 60–64), and all remaining columns take the value 0. Other variables are defined analogously for the remaining retirement trajectories. For example, `h_i_t_61E` contains a value of 40 in the first column (age 60), after which it contains the value zero.

The structure of the variable `y_i_t_65E` mirrors that of `h_i_t_65E`. It contains, in each row, the monthly income at each age from 60 to 100 for the retirement scenario in which early retirement begins at age 65. As illustrated at the bottom of Figure 1, the hypothetical individual earns a monthly wage of €2,000 from ages 60 to 64 and retires at age 65, after which she receives a pension income of €1,400. Accordingly, the first five columns of `y_i_t_65E` take the value €2,000 (for ages 60–64), and all remaining columns take the value €1,400. Other variables are defined analogously for the remaining retirement scenarios.

4.2. Generate Standard-Normal draws using Halton sequences

The second script file, `generate_halton_draws.m`, is used to call the function file `halton.m` as follows:

```
[~,Z] = halton(N_i,halton_N_dimensions,halton_N_draws_per_i, ...
    'prime',halton_primes, ...
    'burn',halton_N_burns, ...
    'leap',halton_N_leaps, ...
    'random',halton_randomize_indicator, ...
    'scramble',halton_scramble_indicator);
```

The `halton` function returns `Z`, a three-dimensional array of size `N_i` $\times$ `halton_N_dimensions` $\times$ `halton_N_draws_per_i`. It contains quasi-random Halton draws transformed to standard normal. The first dimension indexes individuals. The third dimension then provides a sequence of quasi-random draws for each individual, with `halton_N_draws_per_i` draws used to approximate the random-coefficient integral. The second dimension corresponds to the number of prime bases used, which determines the dimensionality of the random-coefficient vector evaluated at each draw.

4.3. Construct the likelihood

The function file `likelihood.m` constructs the objective function. We begin by declaring the standard MATLAB function structure: we specify the output variable (the objective function value), followed by the function name, and lastly its input arguments.

Next, we create the scalar array `N_k`, which specifies the number of covariates included in the model.

The parameter vector `theta` contains all parameters to be estimated. Its first `N_k` elements correspond to the covariate-specific coefficients collected in `beta_l`, while the remaining elements of `theta` are sequentially assigned to scalar parameters.

The array `e_i_l` contains the error terms for the random-coefficient equation. Following equation (13), these errors are generated by scaling standard normal draws by the standard deviation parameter `sigma_l`. Specifically, we set

```
e_i_l(:,1,:) = sigma_l * Z(:,1,:);
```

where `Z` contains standard normal draws. For each individual i and each Halton draw, this operation produces one error term with variance `sigma_l`.

We define the random coefficient using `X_i`, `t`, and the simulated error term `e_i_l` as specified in equation (3):

```
alpha_i_t_l = X_i * beta_l + eta_l * t + e_i_l;
```

where `beta_l` and `eta_l` are parameters to be estimated.

Next, as equation (1) requires, we define the leisure and income terms in logarithmic form. For illustration, we define the leisure term for the early-retirement scenario at the retirement-age regime 65 as

```
log_T_h_i_t_65E = log(T - h_i_t_65E);
```

and we define the income term analogously as

```
log_y_i_t_65E = log(y_i_t_65E);
```

Given these inputs, we are ready to specify the within-period utilities given by equation (2). For illustration, we construct the within-period utility for the early-retirement scenario at the retirement-age regime 65 as

```
U_i_t_65E = alpha_i_t_l .* log_T_h_i_t_65E ...
          + alpha_y .* log_y_i_t_65E ...
          + alpha_l_y .* log_T_h_i_t_65E .* log_y_i_t_65E;
```

The line

```
rho_0_to_40 = rho .^ (t - 60);
```

constructs a vector of discount factors implied by the parameter `rho`, which is to be estimated. The exponent `(t - 60)` indexes the number of years since age 60, so the element corresponding to age t equals ρ^{t-60}. `pi` stores survival probabilities from age 60 onward. Together, these objects provide the discounted survival weights used in computing expected lifetime utility, for example, in the early-retirement scenario at the retirement-age regime 65 as

```
U_i_65E = sum(rho_0_to_40 .* pi .* U_i_t_65E,2);
```

Next, we construct equation (10) as follows:

```
lik_choice_at_age_regime_65 = ...
    sum(exp_U_i_65 .* ind_choice_at_age_regime_65,2) ./ ...
    sum(exp_U_i_65,2);
```

This expression considers the retirement-age regime 65, and gives the probability of choosing a particular scenario by multiplying the indicator for that scenario with the exponentiated utility and then dividing by the total exponentiated utility across all scenarios.

We can now define the joint probability of choosing scenarios at three questions for each i as

```
lik = lik_choice_at_age_regime_61 .* ...
      lik_choice_at_age_regime_63 .* ...
      lik_choice_at_age_regime_65;
```

`lik` is a three-dimensional array of size `N_i` × `halton_N_dimensions` × `halton_N_draws_per_i`, with one entry for each individual and each simulation draw. Recall that the Halton draws index the simulated random-coefficient errors, so the likelihood contribution must be computed separately for each draw. The resulting array corresponds to the product of choice probabilities for a given draw, i.e., the inner term of equation (12) before averaging over simulations.

We average `lik` over the Halton draws for each i:

```
avg_lik = mean(lik,3);
```

Averaging over the third dimension collapses the draw-specific likelihood contributions. Since `lik` contains one likelihood value per individual, per random-coefficient dimension, and per simulation draw, this operation yields a `N_i` × `1` vector representing the simulated likelihood for each individual after integrating over the random-coefficient distribution. This averaging step corresponds exactly to equation (12), which defines the Monte Carlo approximation of the mixed logit likelihood.

The next two steps transform the simulated likelihoods into a form suitable for numerical optimization. First, we take logs:

```
log_avg_lik = log(avg_lik);
```

This operation converts the simulated likelihood for each individual, $\hat{L}_i$ in equation (12), into its log-likelihood contribution ℓ_i defined in equation (14).

We then sum these log-likelihood contributions across individuals and multiply by -1 to obtain the negative sample log-likelihood, which corresponds to $-\ell(\theta)$ in equation (15) and serves as the objective function minimized during estimation:

```
sum_log_avg_lik = -sum(log_avg_lik);
```

4.4. Estimation

The script file, `likelihood_estimation.m`, calls the function file, `likelihood.m` as follows:

```
[theta_hat,fval,exitflag,output,~,grad,hessian] = ...
    fmincon(obj,theta_ig,[],[],[],[],[],[],[],options);
```

Here, `obj` denotes the objective function, which is defined as the negative of the simulated log-likelihood evaluated at a given parameter vector. The vector `theta_ig` contains the initial guesses for the parameters to be estimated. These initial values serve as the starting point for the numerical optimization routine. The function `fmincon` then searches over the parameter space to find the value of `theta` that minimizes the objective function, thereby maximizing the simulated log-likelihood.

4.5. Results

The script file `results_table.m` compiles the parameter estimates, standard error estimates, and performs inference using the corresponding t-statistics. Table 1 presents the results. These estimates can be compared to those reported in Table 5 of Kantarcı et al. (2025). Small differences arise because the present exercise relies on a restricted set of stated-choice questions, whereas the full analysis in Kantarcı et al. uses the complete choice dataset.

Table 1: Estimation results

Parameter	Estimate	Standard error	t value
β^l: constant	-7.013	0.836	-8.393
β^l: age	-0.759	0.171	-4.447
β^l: male	-0.197	0.038	-5.216
β^l: high education	-0.047	0.030	-1.586
β^l: household with no children	0.031	0.044	0.698
β^l: with partner	0.067	0.033	2.028
β^l: home owner	0.042	0.033	1.261
β^l: had a health problem in the last six months	0.053	0.030	1.773
β^l: would work even if money was not needed	-0.082	0.013	-6.467
β^l: experienced or expect early retirement	0.190	0.040	4.749
η^l	0.100	0.010	9.657
σ_l	0.425	0.047	9.056
T	50.191	1.537	32.650
α^y	-0.579	0.158	-3.659
α^{ly}	0.233	0.046	5.023
ρ	1.022	0.015	65.797

5. Final notes

This file is prepared and copyrighted by Hartog Horsch and Tunga Kantarcı. The file and the accompanying MATLAB code are publicly available on GitHub and can be accessed via the following link: link. This exercise draws on three references. The first is: Kantarcı, T., Been, J., van Soest, A., van Vuuren, D. (2025). Partial retirement opportunities and the labor supply of older individuals. Labour Economics, 96, 102739. Available at: link. The second is: Revelt, D., Train, K., 1998. Mixed logit with repeated choices: Households' choices of appliance efficiency level. The Review of Economics and Statistics 80 (4), 647-657. Available at: link. The third is: Gouriéroux, C., Monfort, A., 1990. Simulation based inference in models with heterogeneity. Annals of Economics and Statistics, (20/21), 69-107. Available at: link.